There are others wanting a piece of the action. On the first floor of a business park tower in Nagavara, Bengaluru, Dr Nagendra Nagaraja, CEO and chairman of a startup called QpiAI, has earmarked a corner of his office space for a lab in which an imported dilution refrigerator is due to be installed by September. Having raised $6 million (Rs 50 crore) in funding, Nagaraja is building a 25-qubit quantum computer that he plans to unveil by the end of this year, for which the cryogenic refrigerator is a key component. Once the machine is ready, QpiAI plans to offer the platform to customers via cloud while also exploring a business model of supplying such systems to top institutes and research groups in India. "Our vision is to integrate AI and quantum computing in enterprises," he says, adding that there is an opportunity for Indian companies to be competitive on the world stage. "The more companies you have in advanced technology, the wealthier a nation. It's as simple as that."

In quantum technologies, especially, there's a real opportunity, given that it is still a nascent field globally. This is also where India's Rs 6,000 crore mission assumes significance. The outlay is big, when seen in the historical context of R&D funding in India. But the quantum technology field itself is vast, which means that the funds cannot be spread too thin.

THE AI AND SEMICONDUCTOR CHALLENGE

Meanwhile, the Union cabinet in March this year approved a Rs 10,372 crore mission to strengthen the AI ecosystem in the country. The modalities of the IndiaAI Mission are expected to be announced in a few months. Its objectives include democratising access to computing resources, enhancing data quality, nurturing homegrown AI expertise and promoting socially impactful AI projects. In July, speaking at the Global IndiaAI Summit 2024, Union minister of electronics & information technology Ashwini Vaishnaw said that the government plans to acquire 10,000 graphics processing units (GPUs) as part of the AI mission. "Here, the government will be investing in creating a public platform where compute power, high quality data sets, a common set of protocols, a common framework, technical as well as legal, is available," said Vaishnaw, adding that startups and academicians working on applications for a variety of sectors such as agriculture, medicine, healthcare and education can use this common platform to accelerate their efforts.

In semiconductor manufacturing, India was late to the game even though the country is home to 20-25 per cent of the world's computer chip designers. In a bid to kickstart a manufacturing ecosystem of semiconductors and LCD displays and "de-risk" the country from its dependence on imports, the government, in 2022, had announced a production-linked incentive (PLI) scheme with an outlay of Rs 76,000 crore. In February this year, Tata Electronics announced that it had received government approval to build a mega semiconductor fabrication facility, or fab, in Dholera, Gujarat, in partnership with Taiwan's Powerchip Semiconductor Manufacturing Corporation (PSMC). The fab, into which a total investment of Rs 91,000 crore has been announced, will have a manufacturing capacity of up to 50,000 wafers per month and will generate over 20,000 direct and indirect skilled jobs.

The size of the Indian semiconductor market is expected to reach $109 billion (Rs 9.15 lakh crore) by 2030 from $38 billion (Rs 3.2 lakh crore) in 2023, minister of state for electronics & information technology Jitin Prasada told the Lok Sabha on July 31. The global semiconductor industry is expected to grow to $1 trillion (Rs 83.95 lakh crore) by 2030.

Unlike computer chip manufacturing, there are hardly any commercial fabrication facilities for quantum processors globally, which means it's an opportune time for India to start planning for investments on that front. The four industries likely to see the earliest economic impact from quantum computing would be chemicals, life sciences, finance and mobility, according to the McKinsey report. To illustrate the need for higher investments in emerging technologies, Nagendra Nagaraja of QpiAI picks out two key trends globally: "One is the transition to sustainable energy. And the other is in manufacturing, where the cost of goods has to come down dramatically." According to him, both quantum technologies and AI play a role here by driving innovation. "Material is central to everything," he says. "If we have 300 error-corrected qubits, then I think material science is all disrupted." There's surely a long way to go. But the seeds have to be sown now. ■

TAKEAWAYS

- India has a real opportunity in quantum technologies, given that it is still a nascent field globally. This is where the Rs 6,000 crore National Quantum Mission assumes significance
- The Rs 10,372 crore IndiaAI Mission aims to democratise access to computing resources, enhance data quality, nurture homegrown AI expertise and promote socially impactful AI projects
- In semiconductor manufacturing, India was late to the game, but the size of the domestic market is expected to reach $109 billion (Rs 9.15 lakh crore) by 2030